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BICOL UNIVERSITY - POLANGUI
Polangui, Albay

CPE 26: MICROPROCESSOR

Laboratory Activities:

Compilation of Labs 1 to 7

Submitted By :

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BSCpE – 3A

Submitted to :

Exequiel Mirasol Sabater

Date : March 16, 2026

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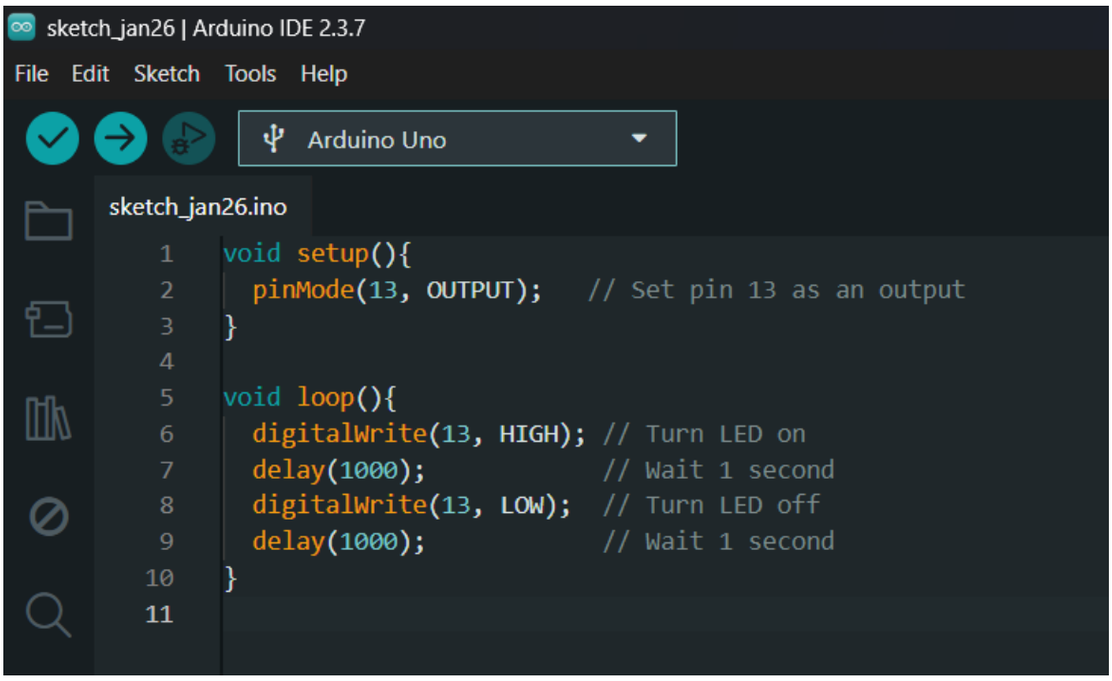
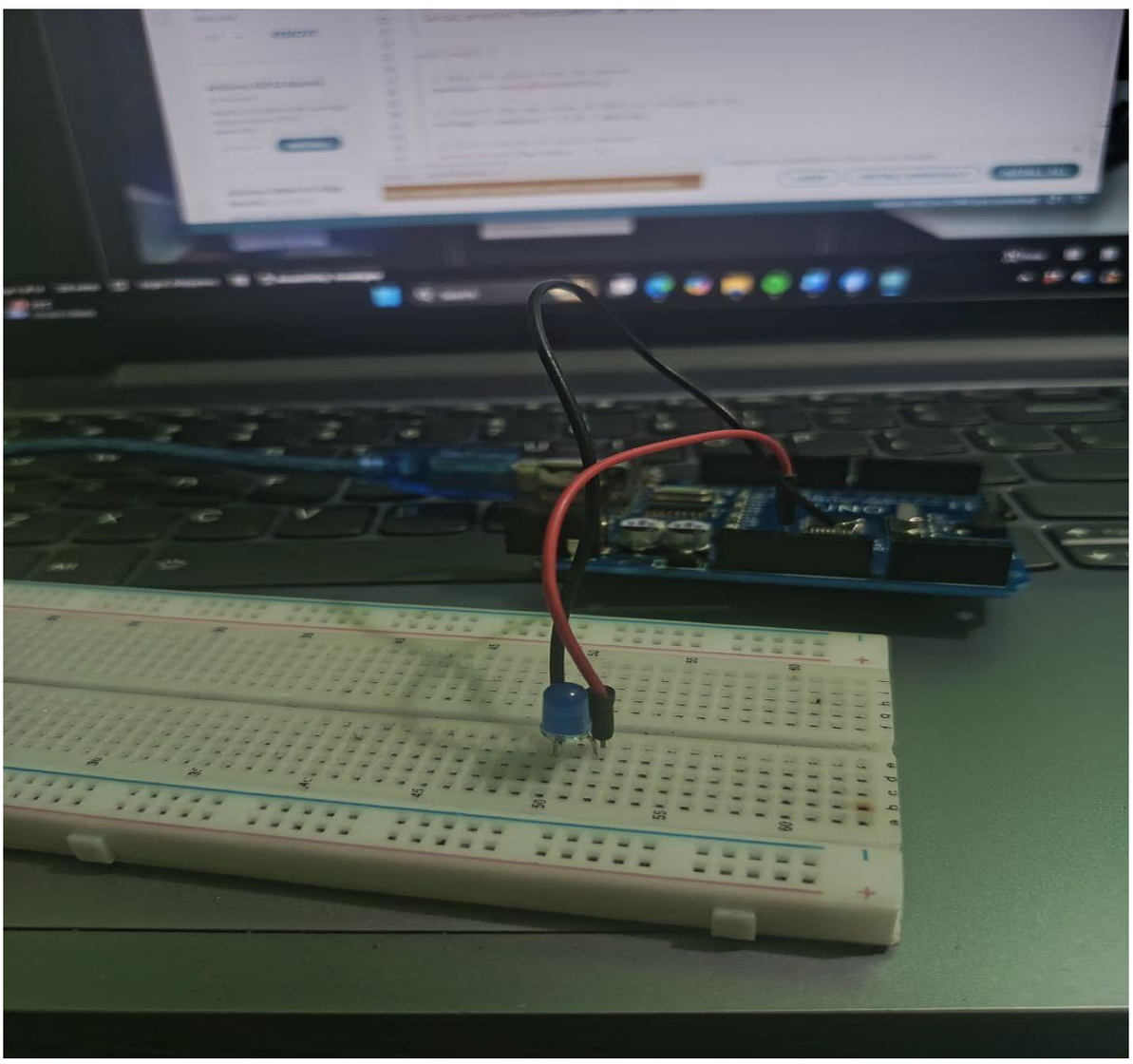
Documentations For Lab 1:





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Documentations For Lab 2:

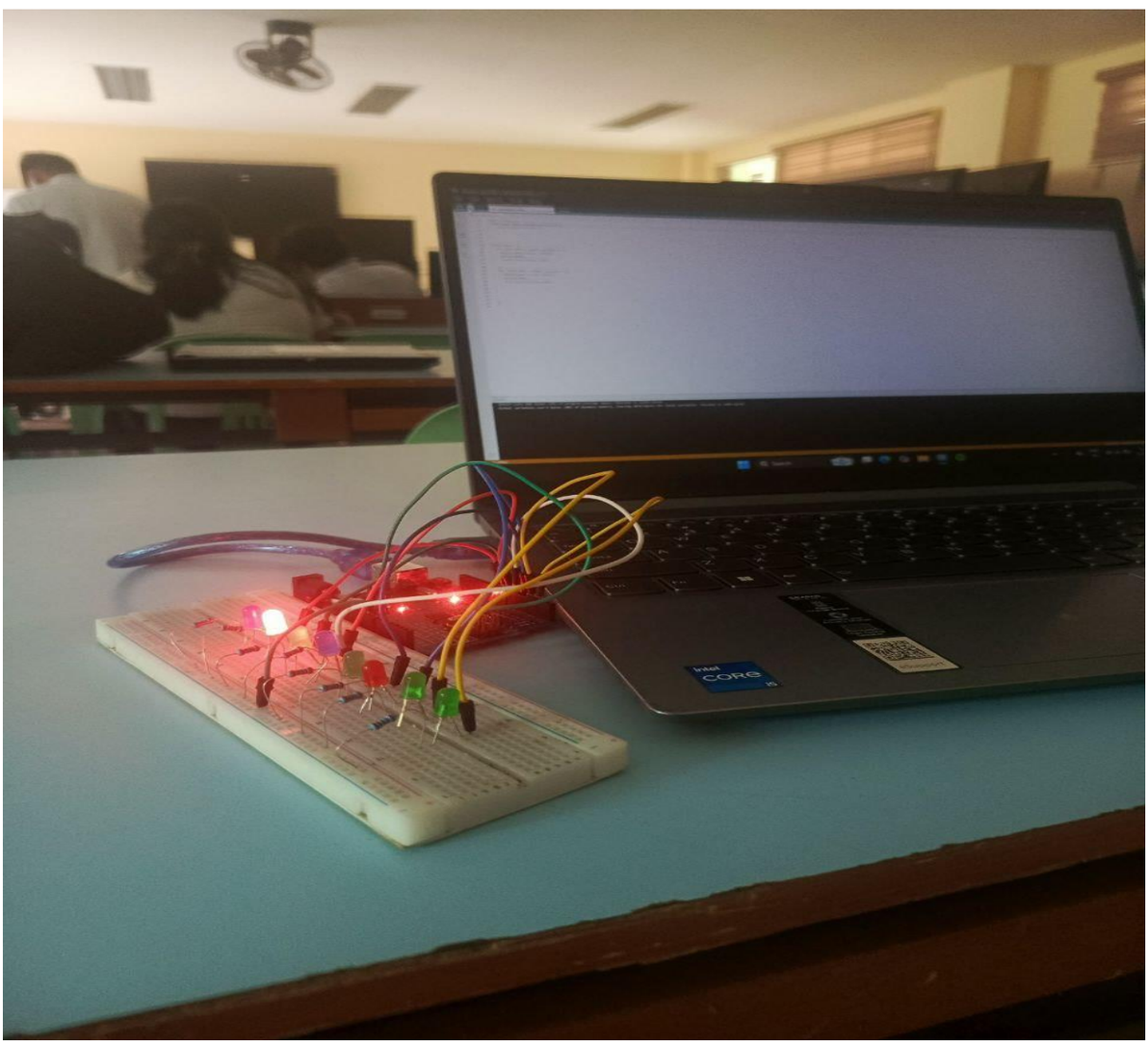
sketch_jan26 | Arduino IDE 2.3.7

File Edit Sketch Tools Help

Arduino Uno

sketch_jan26.ino

```
void setup(){
//Inititalize pins 2 through 9 as OUTPUT using a loop
   for (int pin = 2; pin <=9; pin++){
     pinMode(pin, OUTPUT);
}
}
void loop(){
//Pattern 1: The “Chase” (Left to Right)
    for(int pin = 2; pin<= 9; pin++){
      digitalWrite(pin,HIGH); // Turn LED on
      delay(100);
      digitalWrite(pin,LOW); // Turn LED off
}
//Pattern 2: The “Return” (Right to Left)
    for(int pin = 9; pin >=2; pin--){
      digitalWrite(pin,HIGH);
      delay(100);
      digitalWrite(pin,LOW);
}
}
```



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Documentations For Lab 3:

sketch_jan26 | Arduino IDE 2.3.7

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Arduino Uno

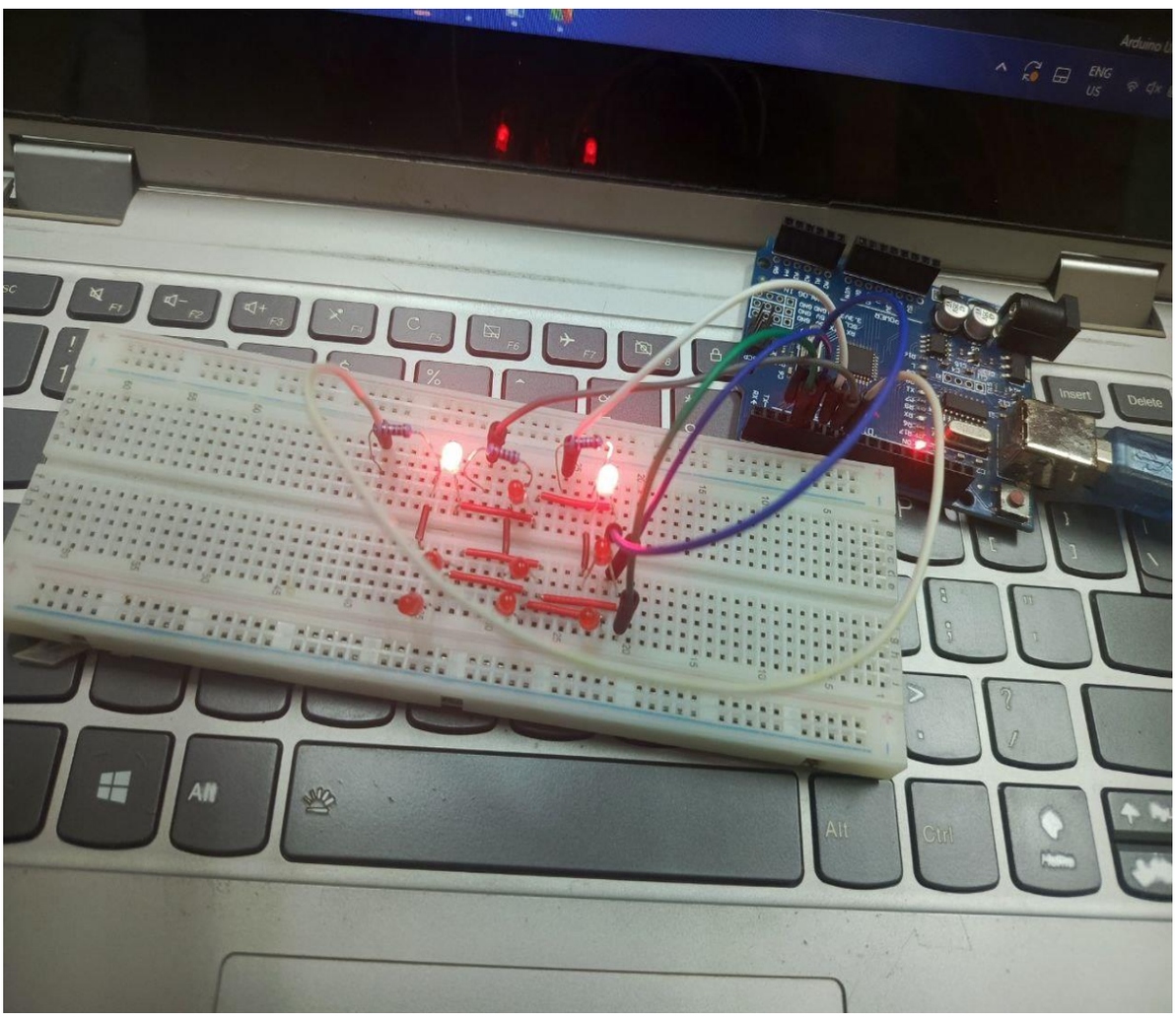
sketch_jan26.ino

```
// Pin definitions for rows and columns
int rowPins[] = {2, 3, 4};   // Rows connected to pins 2, 3, 4
int colPins[] = {5, 6, 7};   // Columns connected to pins 5, 6, 7
// Define the pattern to display on the matrix (1 = LED on, 0 = LED off)
byte pattern[3][3] = {
  {1, 0, 1},  // First row
  {0, 1, 0},  // Second row
  {1, 0, 1}   // Third row
};

void setup() {
  // Set all row pins as output
  for (int i = 0; i < 3; i++) {
    pinMode(rowPins[i], OUTPUT);
  }

  // Set all column pins as output
  for (int i = 0; i < 3; i++) {
    pinMode(colPins[i], OUTPUT);
  }
}
void loop() {
  // Loop through each row and column to display the pattern
  for (int row = 0; row < 3; row++) {
    // Turn on the current row
    digitalWrite(rowPins[row], HIGH);

    for (int col = 0; col < 3; col++) {
      // Turn on the LED in the current row and column if the pattern is 1
      if (pattern[row][col] == 1) {
        digitalWrite(colPins[col], HIGH);
      } else {
        digitalWrite(colPins[col], LOW);
      }
    }
    delay(500); // Delay to make the LED visible
    // Turn off the current row
    digitalWrite(rowPins[row], LOW);
  }
}
```



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Documentations For Lab 4:

```
// LEFT LANE - Christmas Tree Lights
#define LEFT_RED 13
#define LEFT_YELLOW_1 12
#define LEFT_YELLOW_2 11
#define LEFT_YELLOW_3 10
#define LEFT_GREEN 9

// RIGHT LANE - Christmas Tree Lights
#define RIGHT_RED 8
#define RIGHT_YELLOW_1 7
#define RIGHT_YELLOW_2 6
#define RIGHT_YELLOW_3 5
#define RIGHT_GREEN 4

// Ultrasonic Sensor (HC-SR04)
#define TRIG_PIN A0
#define ECHO_PIN A1

// Controls
#define READY_BUTTON A2
#define BUZZER_PIN A3

// Sensor settings
#define MOVEMENT_THRESHOLD 15  // Distance change in cm to detect movement
#define NUM_READINGS 3         // Number of readings to average

// Race States
enum RaceState {
  WAITING_READY,
  COUNTDOWN_DELAY,
  YELLOW_1_PHASE,
  YELLOW_2_PHASE,
  YELLOW_3_PHASE,
  GREEN_PHASE,
  FALSE_START,
  RACE_COMPLETE
};

RaceState currentState = WAITING_READY;
unsigned long stateStartTime = 0;
int baselineDistance = 0;

void setup() {
  Serial.begin(9600);

  // Configure LED pins
  pinMode(LEFT_RED, OUTPUT);
  pinMode(LEFT_YELLOW_1, OUTPUT);
  pinMode(LEFT_YELLOW_2, OUTPUT);
  pinMode(LEFT_YELLOW_3, OUTPUT);
  pinMode(LEFT_GREEN, OUTPUT);

  pinMode(RIGHT_RED, OUTPUT);
  pinMode(RIGHT_YELLOW_1, OUTPUT);
  pinMode(RIGHT_YELLOW_2, OUTPUT);
  pinMode(RIGHT_YELLOW_3, OUTPUT);
  pinMode(RIGHT_GREEN, OUTPUT);

  // Configure ultrasonic sensor
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  // Configure controls
  pinMode(READY_BUTTON, INPUT_PULLUP);
  pinMode(BUZZER_PIN, OUTPUT);

  // Initialize
  turnOffAllLights();
  digitalWrite(BUZZER_PIN, LOW);

  // Show RED light
  digitalWrite(LEFT_RED, HIGH);
  digitalWrite(RIGHT_RED, HIGH);

  Serial.println(F("\n=== DRAG RACE READY ===="));
  Serial.println(F("Single Sensor - Improved Stability"));
  Serial.println(F("RED LIGHT ON - Press button\n"));
}
```

```
void loop() {
  switch (currentState) {
    case WAITING_READY:
      handleWaitingReady();
      break;
    case COUNTDOWN_DELAY:
      handleCountdownDelay();
      break;
    case YELLOW_1_PHASE:
      handleYellow1();
      break;
    case YELLOW_2_PHASE:
      handleYellow2();
      break;
    case YELLOW_3_PHASE:
      handleYellow3();
      break;
    case GREEN_PHASE:
      handleGreen();
      break;
    case FALSE_START:
      handleFalseStart();
      break;
    case RACE_COMPLETE:
      handleComplete();
      break;
  }
}

void handleWaitingReady() {
  if (digitalRead(READY_BUTTON) == LOW) {
    delay(50);
    if (digitalRead(READY_BUTTON) == LOW) {
      Serial.println(F("\n=== BUTTON PRESSED ===")); 

      // Take multiple readings and average them
      Serial.println(F("Calibrating sensor...")); 
      delay(200);  // Let sensor stabilize

      baselineDistance = getStableDistance();

      Serial.print(F("Baseline: ")); 
      Serial.print(baselineDistance);
      Serial.println(F(" cm")); 

      if (baselineDistance < 5 || baselineDistance > 300) {
        Serial.println(F("WARNING: Sensor reading unusual!")); 
        Serial.println(F("Check sensor position")); 
      }

      tone(BUZZER_PIN, 800, 500);
      changeState(COUNTDOWN_DELAY);
    }
  }
}

void handleCountdownDelay() {
  if (millis() - stateStartTime >= 1000) {
    changeState(YELLOW_1_PHASE);
  }
}

void handleYellow1() {
  if (checkMotionDuringStaging()) return;

  if (millis() - stateStartTime >= 1000) {
    changeState(YELLOW_2_PHASE);
  }
}

void handleYellow2() {
  if (checkMotionDuringStaging()) return;

  if (millis() - stateStartTime >= 1000) {
    changeState(YELLOW_3_PHASE);
  }
}
```

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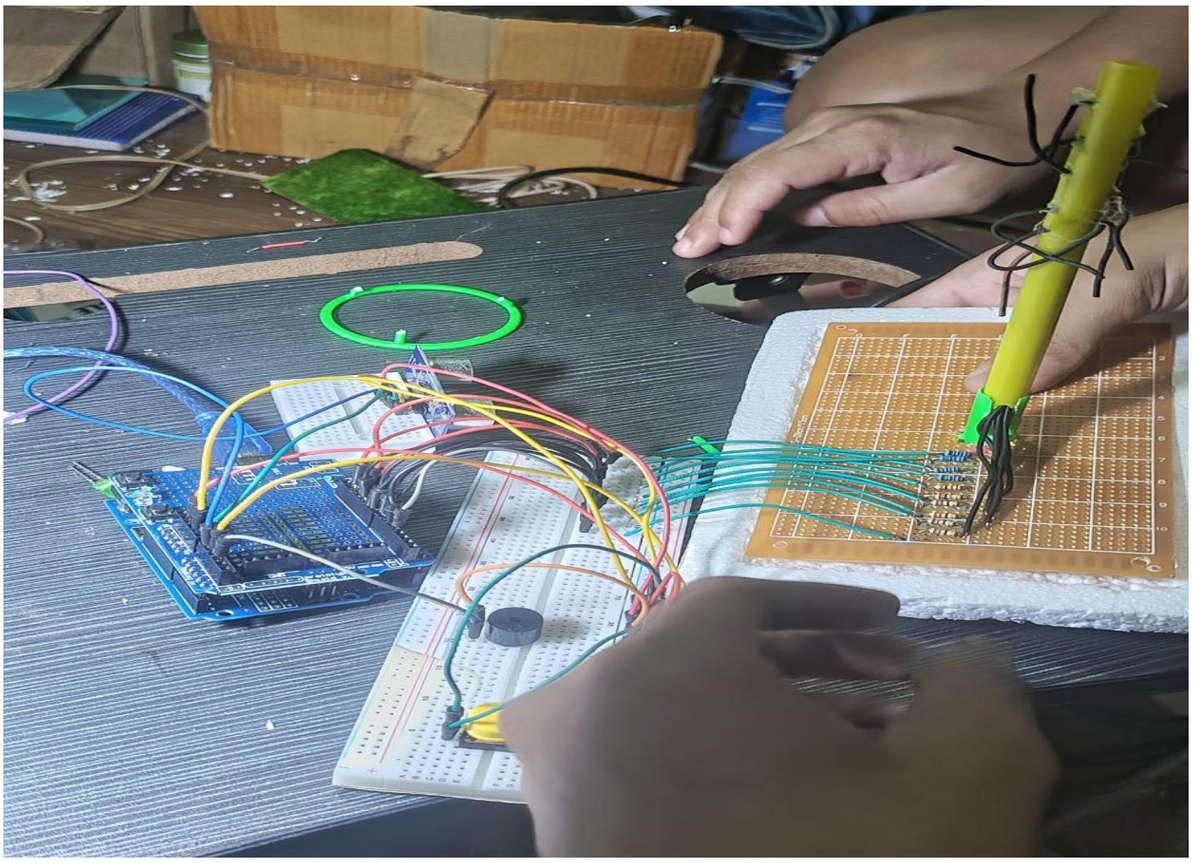
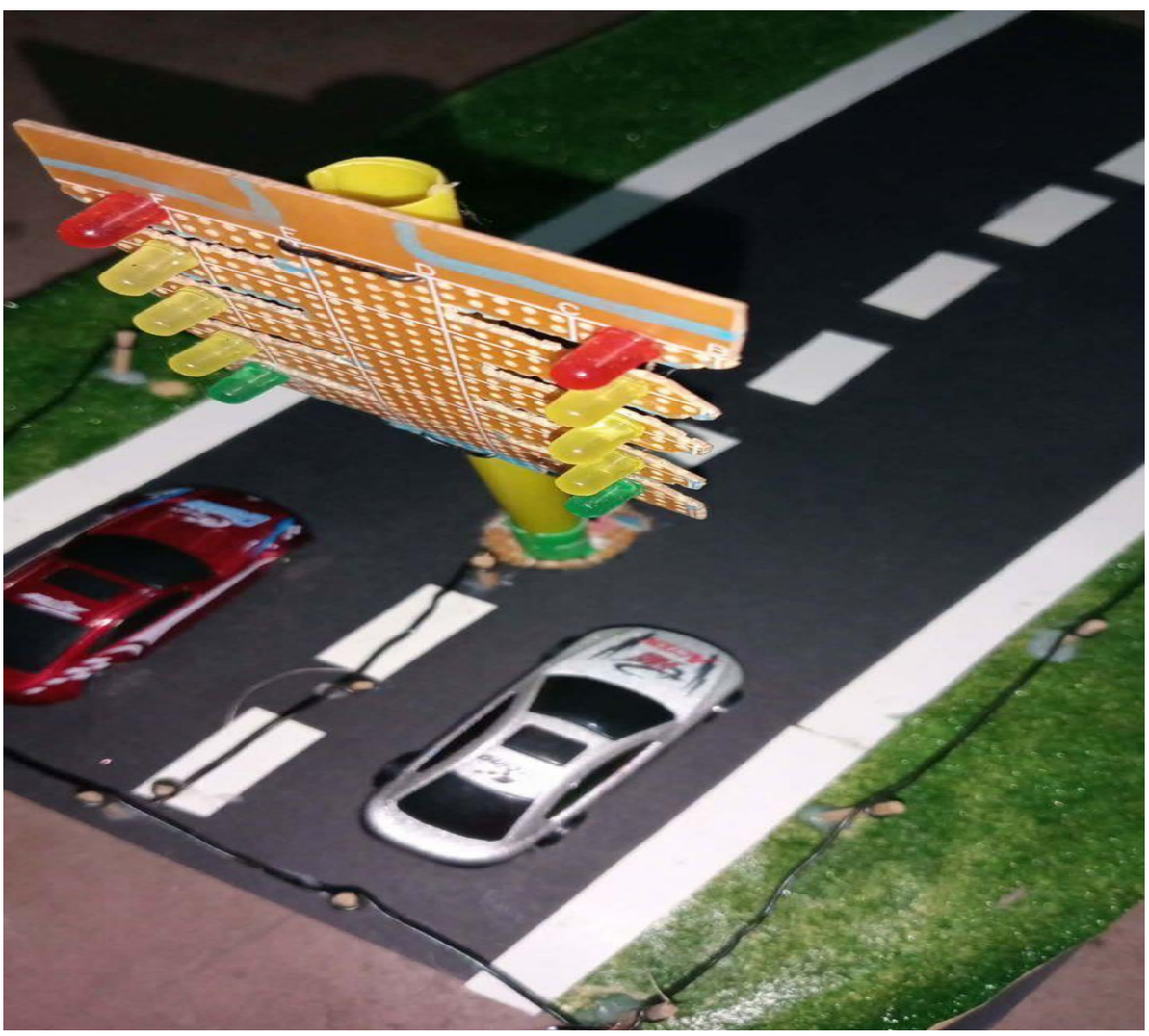
```
  if (distanceChange > MOVEMENT_THRESHOLD) {
    Serial.println(F("\n*** EARLY MOVEMENT ***"));
    changeState(FALSE_START);
    return true;
  }

  return false;
}

int getStableDistance() {
  // Take multiple readings and average them for stability
  long total = 0;
  int validReadings = 0;

  for (int i = 0; i < NUM_READINGS; i++) {
    int reading = getSingleDistance();

    // Filter out bad readings (0 or > 300cm)
    if (reading > 0 && reading < 300) {
      total += reading;
      validReadings++;
    }

    delay(30);  // Small delay between readings
  }

  if (validReadings == 0) {
    return 400;  // No valid readings
  }

  return total / validReadings;  // Return average
}

int getSingleDistance() {
  // Send pulse
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  // Read echo with timeout
  long duration = pulseIn(ECHO_PIN, HIGH, 30000);

  // Calculate distance
  if (duration == 0) return 0;  // No echo

  int distance = duration * 0.034 / 2;
  return distance;
}

void turnOffAllLights() {
  digitalWrite(LEFT_RED, LOW);
  digitalWrite(LEFT_YELLOW_1, LOW);
  digitalWrite(LEFT_YELLOW_2, LOW);
  digitalWrite(LEFT_YELLOW_3, LOW);
  digitalWrite(LEFT_GREEN, LOW);

  digitalWrite(RIGHT_RED, LOW);
  digitalWrite(RIGHT_YELLOW_1, LOW);
  digitalWrite(RIGHT_YELLOW_2, LOW);
  digitalWrite(RIGHT_YELLOW_3, LOW);
  digitalWrite(RIGHT_GREEN, LOW);
}

void setAllLights(int state) {
  digitalWrite(LEFT_RED, state);
  digitalWrite(LEFT_YELLOW_1, state);
  digitalWrite(LEFT_YELLOW_2, state);
  digitalWrite(LEFT_YELLOW_3, state);
  digitalWrite(LEFT_GREEN, state);

  digitalWrite(RIGHT_RED, state);
  digitalWrite(RIGHT_YELLOW_1, state);
  digitalWrite(RIGHT_YELLOW_2, state);
  digitalWrite(RIGHT_YELLOW_3, state);
  digitalWrite(RIGHT_GREEN, state);
}
```





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Documentations For Lab 5:

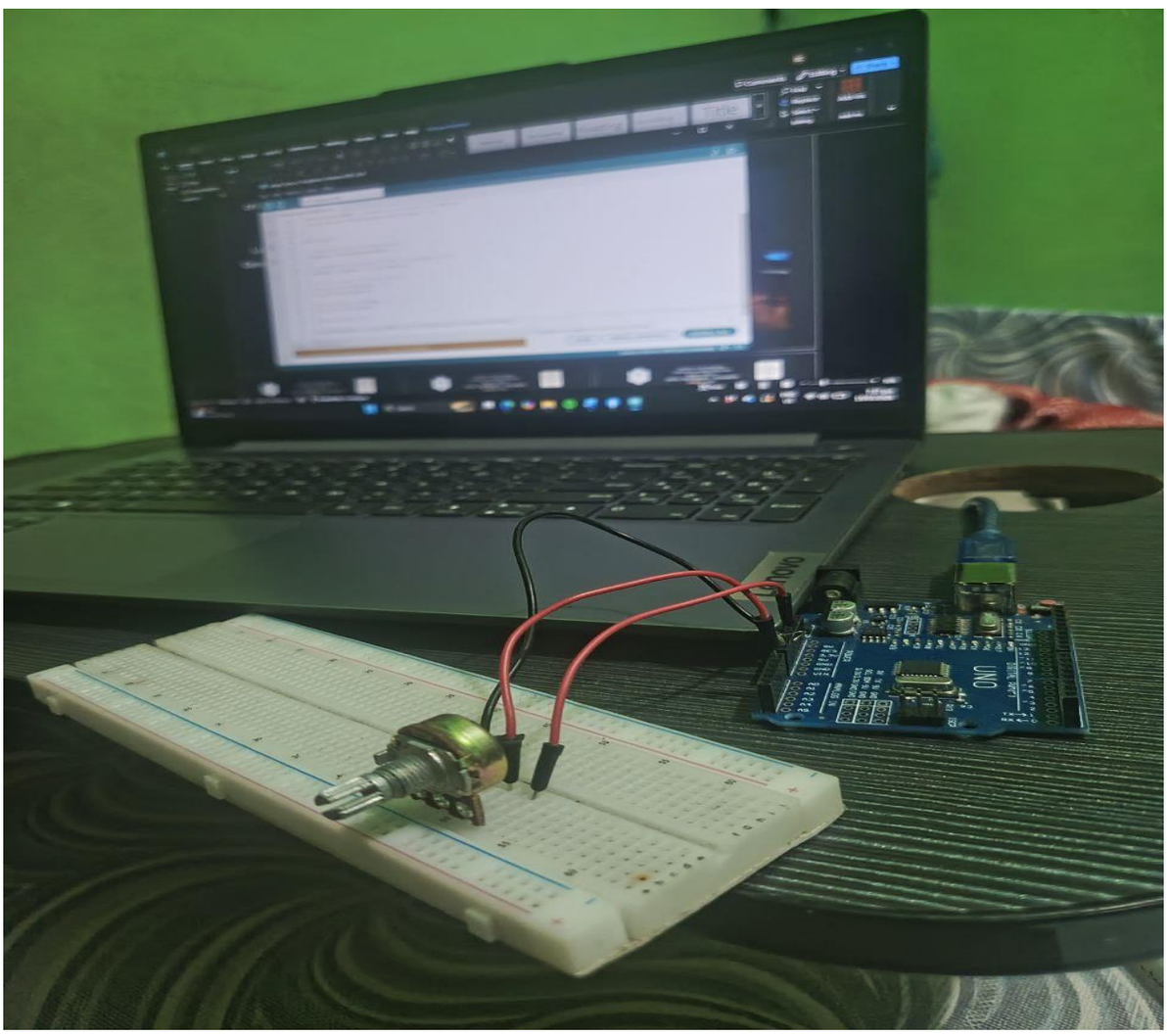
sketch_jan26 | Arduino IDE 2.3.7

File Edit Sketch Tools Help

Arduino Uno

sketch_jan26.ino

```
/*
  Analog Input Lab
  Reads an analog input on pin 0, prints the result to the Serial Monitor.
*/
int potPin = A0;      // The middle pin of the potentiometer
int rawValue = 0;     // Variable to store the raw ADC value (0-1023)
float voltage = 0;    // Variable to store the calculated voltage

void setup() {
  Serial.begin(9600); // Open the serial port at 9600 bps
  Serial.println("Potentiometer Lab Started...");
}

void loop() {

  // Read the value from the sensor
  rawValue = analogRead(potPin);

  // Convert the raw value (0-1023) to voltage (0-5V)
  voltage = rawValue * (5.0 / 1023.0);

  // Print results to Serial Monitor
  Serial.print("Raw Value: ");
  Serial.print(rawValue);
  Serial.print(" | Voltage: ");
  Serial.print(voltage);
  Serial.println("V");

  delay(200); // Wait 200ms so the screen doesn't flicker too fast
}
```



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Documentations For Lab 6:

sketch_jan26 | Arduino IDE 2.3.7

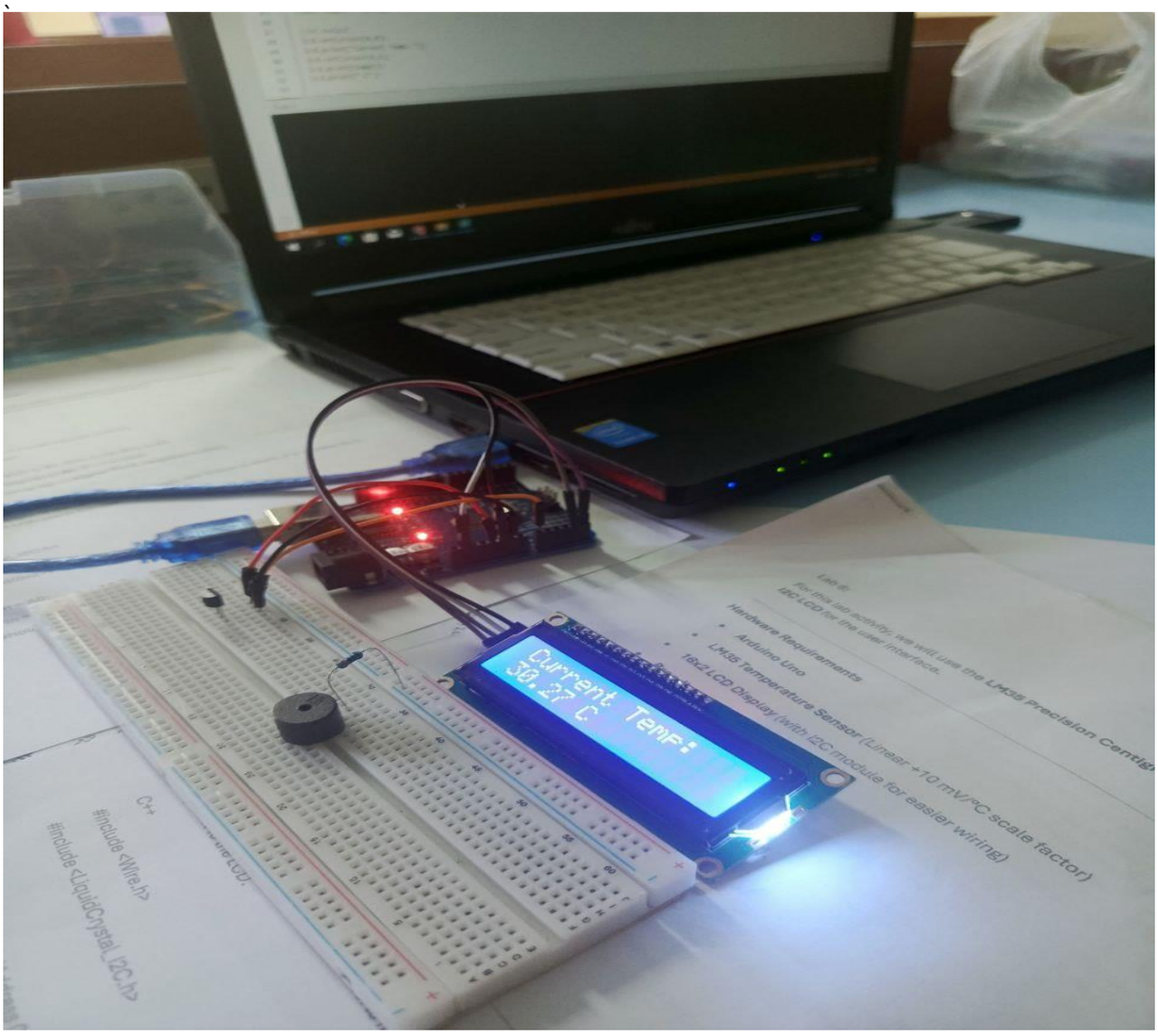
File Edit Sketch Tools Help

Arduino Uno

sketch_jan26.ino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Initialize the LCD (Address 0x27 is common)
LiquidCrystal_I2C lcd(0x27, 16, 2);

const int sensorPin = A0;

void setup() {

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("Temp Monitor");
  delay(2000);

  lcd.clear();
}

void loop() {

  // Read analog value (0-1023)
  int reading = analogRead(sensorPin);

  // Convert reading to voltage (for 5V Arduino)
  float voltage = reading * (5.0 / 1024.0);

  // LM35 output is 10mV per degree Celsius
  float tempC = voltage * 100.0;

  // UI Output
  lcd.setCursor(0, 0);
  lcd.print("Current Temp:");

  lcd.setCursor(0, 1);
  lcd.print(tempC);
  lcd.print(" C");

  delay(1000); // Update every second
}
```

`



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Documentations For Lab 7:

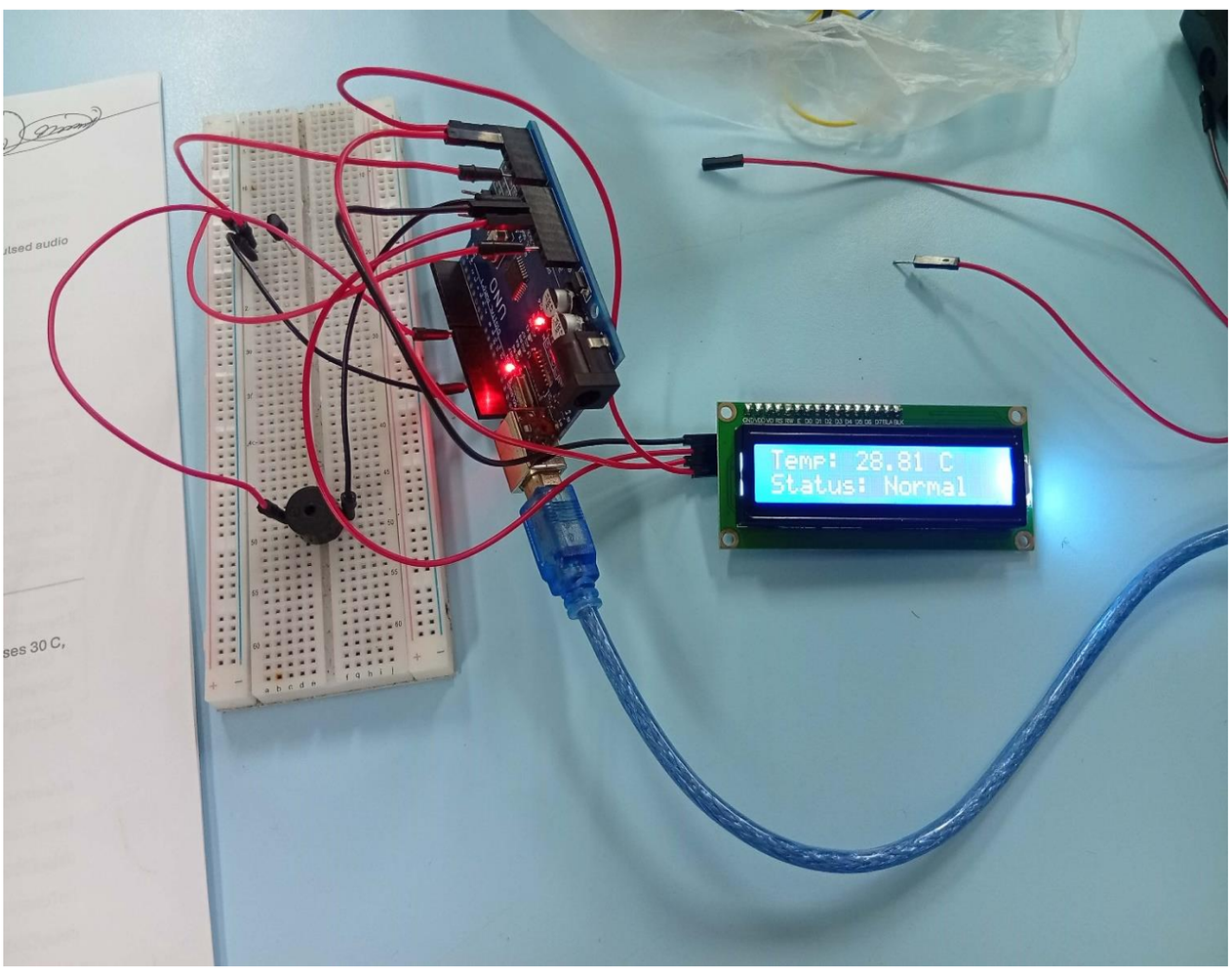
sketch_jan26 | Arduino IDE 2.3.7

File Edit Sketch Tools Help

Arduino Uno

sketch_jan26.ino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

const int sensorPin = A0;
const int buzzerPin = 8;
const float threshold = 30.0; // Trigger temp in Celsius

void setup() {
  lcd.init();
  lcd.backlight();
  pinMode(buzzerPin, OUTPUT);
}

void loop() {

  int reading = analogRead(sensorPin);

  float voltage = reading * (5.0 / 1024.0);

  float tempC = voltage * 100.0;

  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(tempC);
```



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CpE Elec 2: COGNATE 2

Laboratory Activities:

Lab 1 and 2

Submitted By :

Gian Carlo D. Trilles

BSCpE – 3A

Submitted to :

Exequiel Mirasol Sabater

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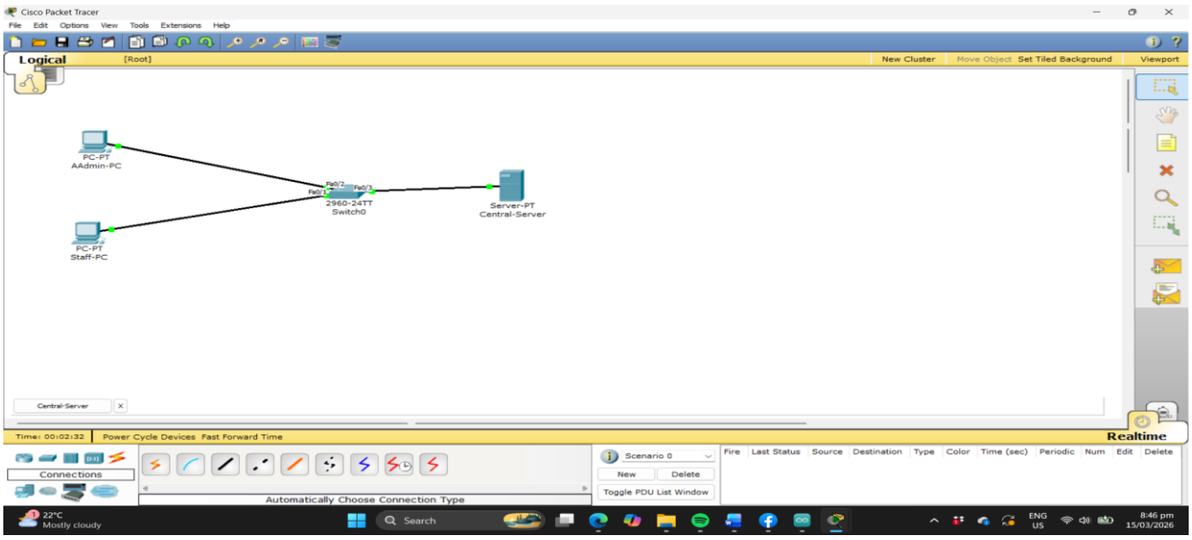
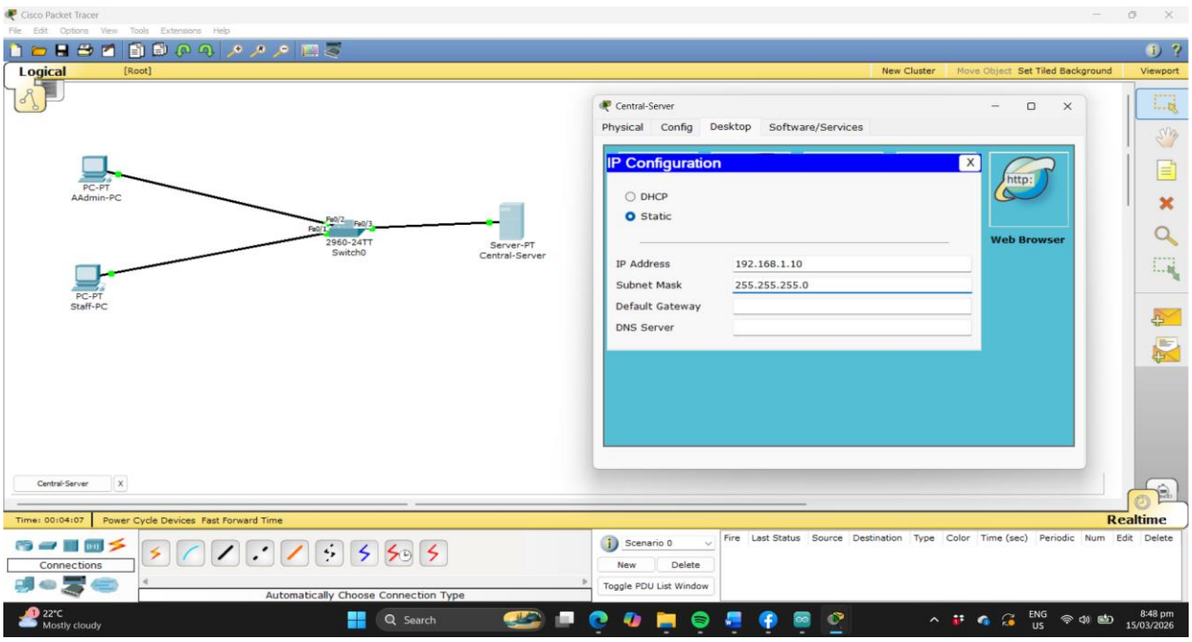
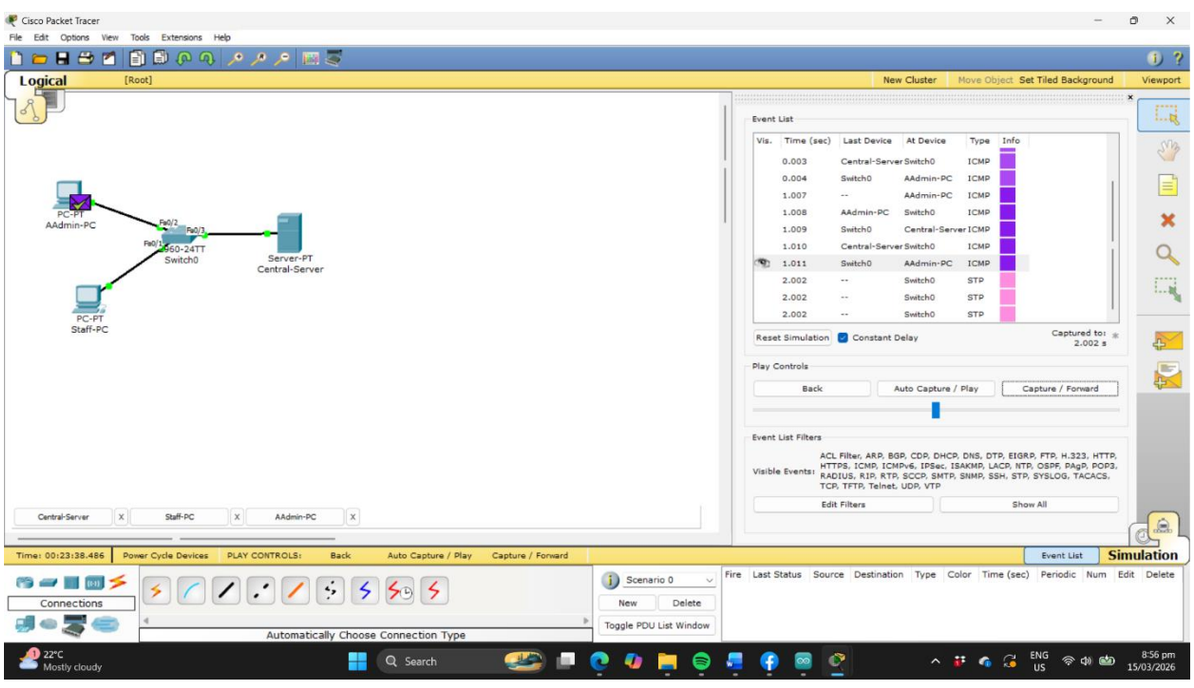
Documentations for Lab 1: Phase1



Phase2:



Phase3



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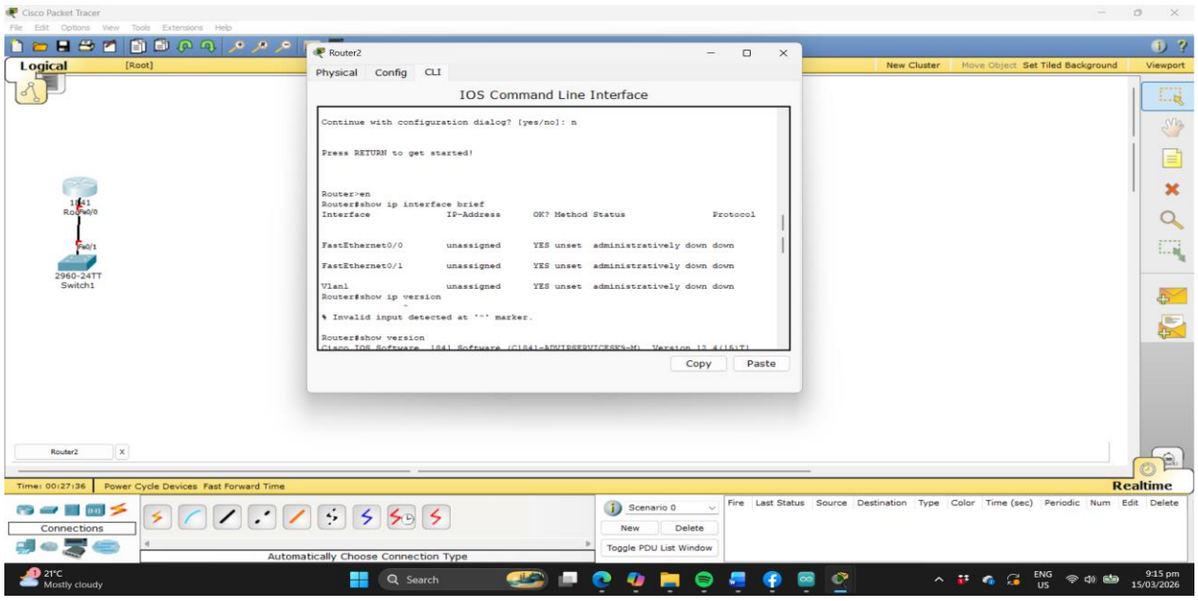
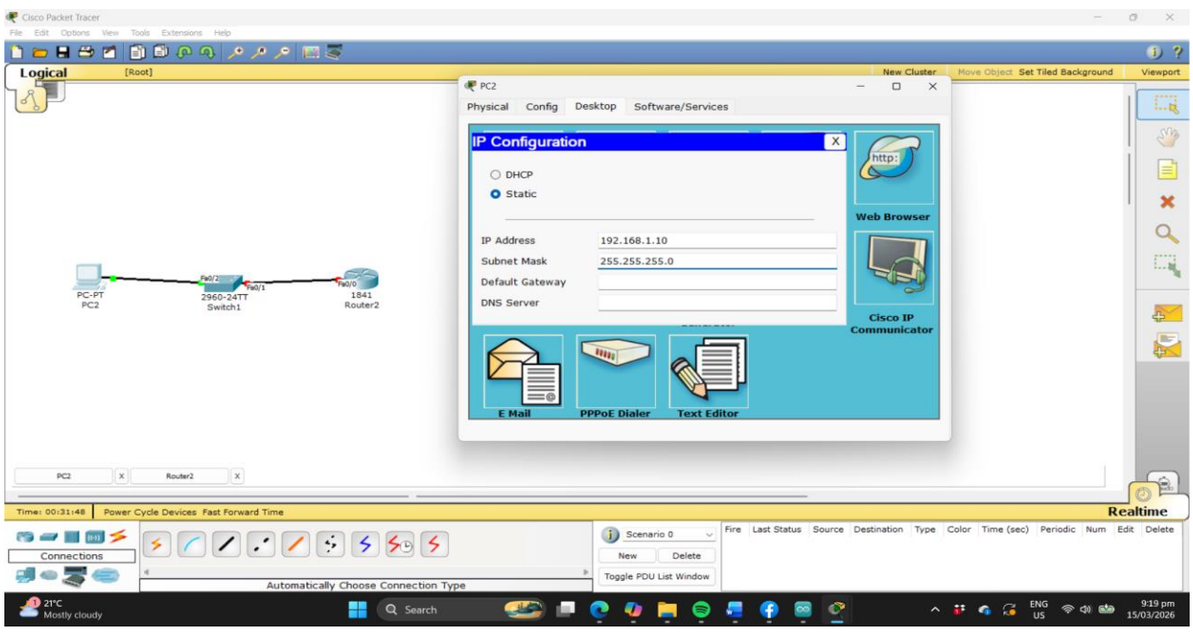
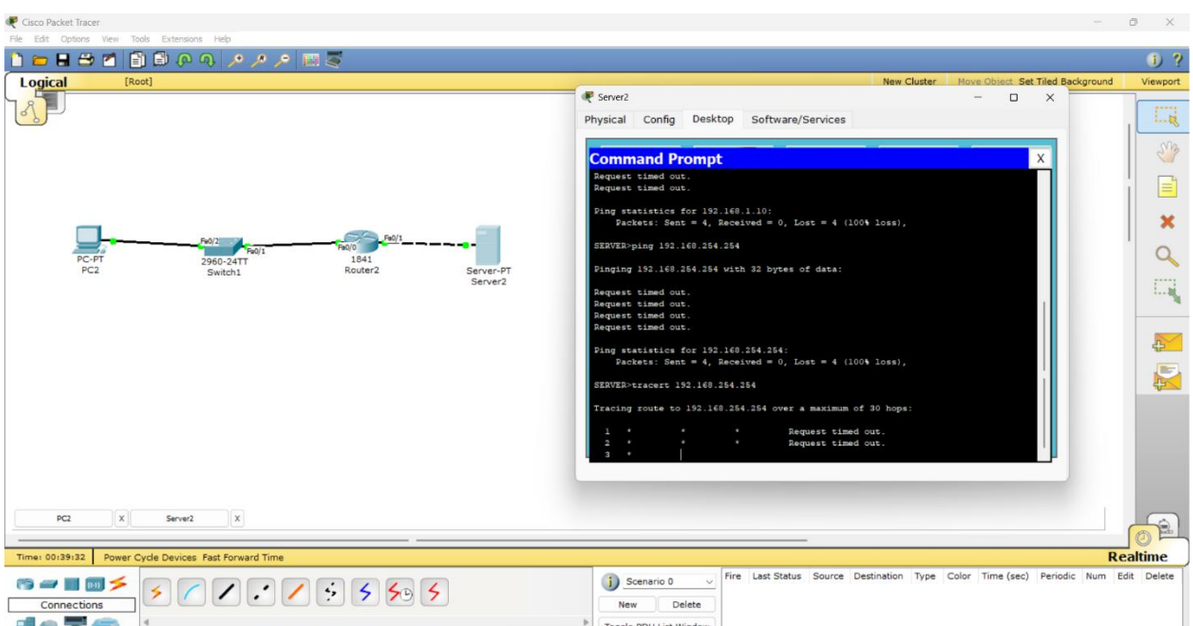
Documentations for Lab 2: Part1



Windows environment



Linux environment